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# **NUTRITION AND METABOLISM**

# NG tube insertion

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- Passing a tube through a nasal cavity down the nasopharynx and oesophagus into the stomach

## Purpose

- To administer tube feeding and medication to clients unable to eat by mouth or swallow a sufficient diet without aspirating food or fluids into the lungs.
- To establish a means for suctioning stomach contents to prevent gastric distension, before and after surgery, nausea and vomiting
- To remove stomach contents for laboratory analysis
- To decompress abdominal distension.
- To lavage (wash) the stomach in case of poisoning or overdose of medications

# Indication

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- ☐ Surgery
- ☐ Abdominal distension
- ☐ Poison
- ☐ Unconscious
- ☐ Severe dehydration
- ☐ Diagnostic analysis

# Position During NGT insertion

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- Conscious pt:
  - Semi sitting position
- Unconscious pt:
  - Left lateral position

## ❑ **Contraindication of NGT Insertion**

- Basilar skull fracture
- Recent nasal, esophageal or gastric surgery
- Esophageal stricture
- Esophageal varices
- Bleeding diathesis
- Hx of caustic/acid ingestion or esophagus burns

## Complications

- Nasotracheal intubation (malposition)
- Esophageal perforation
- Esophagitis, sinusitis
- Pulmonary aspiration
- GI bleeding
- Intracranial placement of the tube

# Precaution During NGT Insertion

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- ***Check the position of the tube to confirm that it is in the stomach by:-***
  - Introducing 10-20ml of air in to the stomach via the tube and check for a **whooshing/gurgling sound** using a stethoscope placed over the epigastrium
  - Aspirating the contents of the stomach with a syringe. The aspirate turns **blue litmus paper to red**, due to HCl.
  - Insert/immerse the tip of tube into water in the kidney dish and if you see gentle and continuous bubble that show you are in the lung

# Gastric Gavage/NGT Feeding

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- Is providing nutritional supplement when the patient is unable or not willing to take food per mouth with normal GI tract functioning
- Feeding of a client with NGT when ever the pt unable or un willing to take adequate amount of food per oral.
- Feeding of unconscious pt those who have laryngeal or oral operation by NGT

# NGT feeding....

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## **Purpose:**

- To provide total supplemental nutrition
- Restore fluid, electrolyte and acid base balance.
- Reduce or eliminate catabolism and negative nitrogen balance
- To provide food for pt with paralysis of throat  
fractured jaw premature baby unable to suck breast

## **Precaution**

- Severe pancreatitis, Enterocutaneous fistulae
- GI ischemia

## ➤ **Contraindication**

- ❑ Diffused peritonitis
- ❑ Intestinal obstruction that prohibits normal bowel functioning
- ❑ Intractable vomiting, paralytic ileus
- ❑ Severe diarrhea

# NGT feeding...

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## Complications:

- **Diarrhoea:** due to hyper osmolar feeding, rapid infusion, bacteria contaminated feedings, lactase deficiency and food allergies etc.
- **Nausea/vomiting:** due to Change in rate of feeding, offensive smell, inadequate gastric emptying
- **Cramping/gas:** due to air in tube
- **Constipation:** high milk content, low fiber intake, inadequate fluid intake.

# NGT feeding...

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- **Aspiration pneumonia:** due to improper tube placement, flat in bed, too large tube etc.
- **Tube displacement:** due to excessive coughing/vomiting, tracheal suctioning etc.
- **Tube obstruction:** due to inadequate flushing/formula rate
- **Nasopharyngeal irritation:** due to tube position and large tubes
- **Hyperglycemia:** glucose intolerance and high carbohydrate feeding content

- Aspirate stomach contents to determine amount of residual and measure it.
  - If the residual is over 100 ml in adults and 10 ml or more infants, hold the feeding until residual diminishes or subtract the withdrawn amount from the total feeding and administer the rest
  - All these are based on the policy agency.

# NGT feeding...

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## **N.B.**

- ❑ After feeding introduce 50 to 60 ml of water, to prevent tube syndrome & further blockage.
- ❑ Deliver feeding over the desired length of time. Usually 200-350 ml over 10-15 minutes is given the patient should be installing-up position if conscious
- ❑ The nostril should be cleaned with clean water or sodium bicarbonate.
- ❑ Mackintosh with towel are placed around the neck chest of the pt
- ❑ Lubricate the tube with cold water insert in to one nostril with down ward direction.
- ❑ If the pt is not deeply unconscious tell him to swallow-see the tube not come forward &coiled up in the mouth
- ❑ Ring the tube 40 to 45cm in to the stomach

# NGT feeding....

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## Equipment

- ❑ Trolley
- ❑ Food (liquid diet at room temperature) (370to380) or glass of in bowel with warm water
- ❑ NGT, syringe or gavage beg with flow regulatory clamp
- ❑ Towel(mackintosh), napkin
- ❑ 50ml syringe
- ❑ IV stand adhesive tape
- ❑ K-Y jelly
- ❑ Sodium bicarbonate or clean water in gallipots to clean the nostril with wooden applicators.
- ❑ Clean cold water in kidney dish to lubricate the tube or to check up the direction of tube
- ❑ Spigot (if the tube is retaining type)
- ❑ Receiver for used equipment & swabs

# NGT feeding...

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## Procedure

- ☐ Explain the procedures to the client
- ☐ wash your hands & dry
- ☐ Clean the trolley
- ☐ Assemble all the necessary material to the bed side
- ☐ keep the pt privacy, cover the pt chest with towel or mackintosh
- ☐ Insert the NGT with appropriate procedure
- ☐ Connect the gavage bag with NGT & open the clamp or give slowly with syringe
- ☐ Never allow the syringe or gavage bag to empty completely to prevent entrance of air

# NGT feeding....

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- Usually provide 200 to 350 ml over 10 to 15 minutes
- Flush the tubing by adding about 60 ml of water to the gavage bag or syringe to maintain tube from blocking after demonstration of the food
- If you are administering a continues feeding flush the feeding tube Q4 hours to prevent tube occlusion
- When you are D/C gavage, cover the end the of the tube with its plug or cap to prevent leakage &contamination of the tube
- Leave the pt in semi fowlers or high fowlers position for at least 30 minutes.
- record the procedure, in &out put& general statues of the patient

# GASTROSTOMY FEEDING

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- Surgical insertions of tube in to the stomach, for feeding or introduce medication.
- Giving Medication or Nutrition through gastrostomy tube
- **Indication**

# Gastrostomy feeding....

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## Training Material

- Funnel, rubber tubing, glass connector, clamp (sterile)
- Glass of drinking water.
- Required amount of food (37-38°C)
- Sterile lubricant to protect surrounding area
- Dressing set (2 forceps & 1 scissor).
- Medication ordered.
- Kidney tray
- Marry tailed binder if required.
- Mackintosh & towel.
- Stethoscope, syringe.

# Gastrostomy feeding....

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## ***Procedure:***

- Explain the procedure to the pt
- Wash your hands
- Assemble all the necessary equipment & take them to the bed side
- Screen the pt
- Place the mackintosh or towel under the pt, & clean the surrounding area & cover the area with piece of gauze.
- Attach funnel with the tube, but pinch the rubber tubing to prevent entrance of air, introduce the food with jug

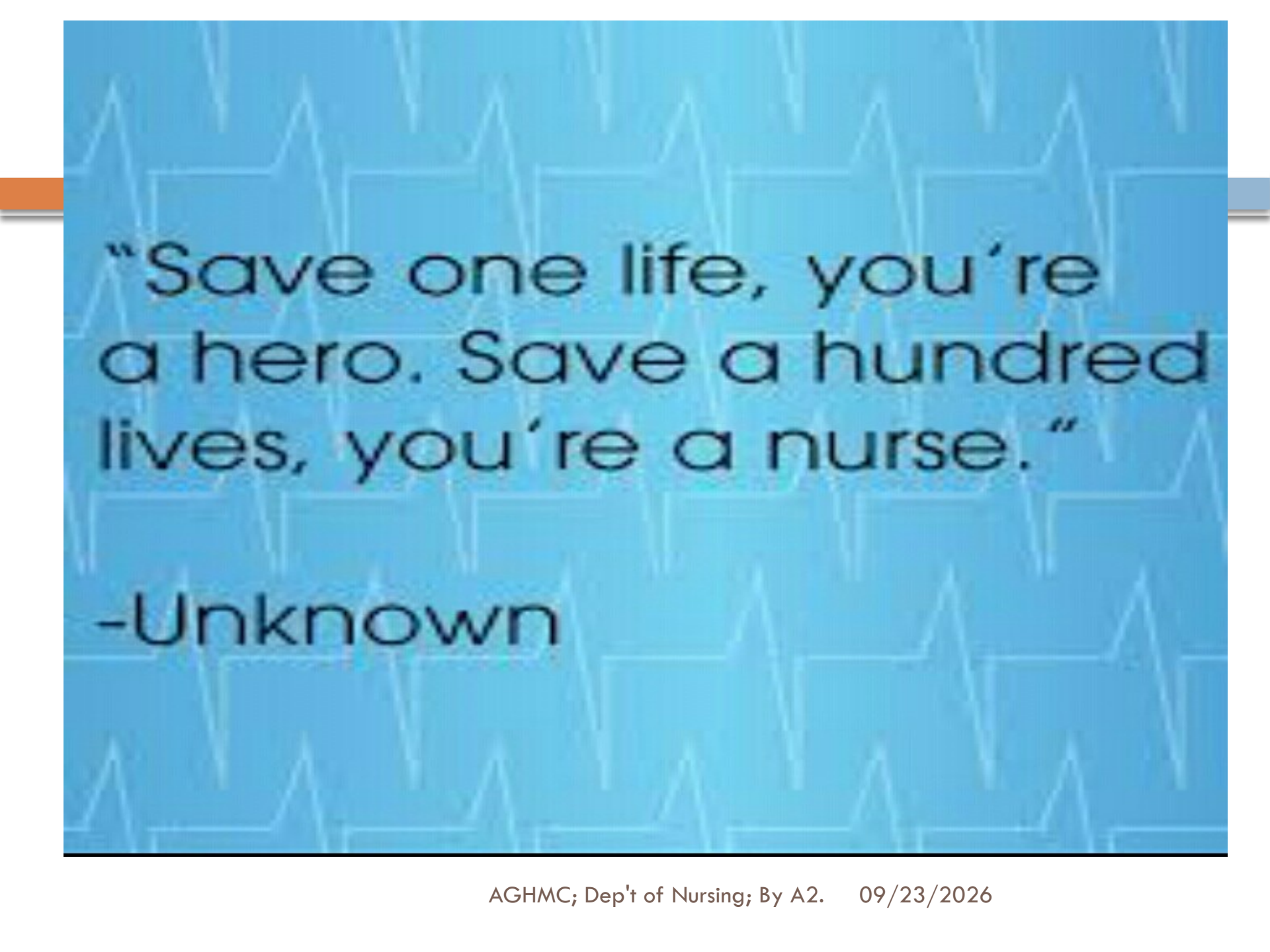
# Gastrostomy feeding....

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- Pour some clear water in to the tube to know the potency of the tube
- Then pour the food before the funnel is empty
- If any medications are ordered give accordingly (after the food)
- Provide water after medication or food
- Disconnect the tube & funnel
- Clean, and apply sterile dressing around the stoma apply binder
- Clean, & boil or sterile the equipment & return in their proper places, record the type of food, amount & time.
- Report any discomfort or condition to the pt.

# Thank you!!!

*Be in act as you have  
been in thought!!*



“Save one life, you’re  
a hero. Save a hundred  
lives, you’re a nurse.”

-Unknown